

DS8003 Final Project Description

Goal: To perform some data analysis using the different big data tools and report results

Requirements:

1. To use the tools learnt in at least 4/10 lectures in this course in the project
2. Find a problem definition that makes sense for the dataset you have chosen
3. Produce 5 insights or findings from the data that relate to the problem definition. You are welcome to use visualizations or statistics or other forms of reporting to showcase your results
4. Each of the 5 insights should have used one or more of the tools that were taught in this course.
5. You are welcome to use additional tools that are outside of the scope of the course, but it does not count towards the 4/10 requirement.
6. We have provided some sample datasets with some ideas, but you are welcome to choose another dataset entirely (provided it is in a similar scale to what's been provided as samples).
7. You are expected to work in groups of 4 people in the project.
8. You are supposed to submit the following at specified times (check the course outline and D2L Assignment Page for due dates. Check the Rubric below for the format and content of the documents):
 - a. Project Proposal
 - b. Final Presentation
 - c. Project Document
9. Put your group information as well as the selected topic on the spreadsheet on this link:
<https://docs.google.com/spreadsheets/d/16dkZ2xW56INtMbjZepPStDro0D3G-AIKN1xr-PapE8E/edit?usp=sharing>

Rubric:

1. Project Proposal (1-2 pages) (5 points)
 - a. Problem Definition - 2 points
 - b. Suggested solution (focus on how it will satisfy the requirements) - 2 points
 - c. Dataset descriptions - describe some of the key attributes you want to potentially use in your analysis and open-source link to download the dataset - 1 point
2. **Project Document (at most 20 pages; 12 point and 1.5 line spacing; Calibri) (25 points)**
 - a. Problem Definition - 1 points
 - b. Data Description - 2.5 points
 - i. Attributes description - 0.5 point

- ii. Statistics of the data (use the tools learnt in this course to generate the data statistics) - 2 points
 - c. Work distribution
 - d. Solution description - 13 points (3.25 points per tool usage)
 - i. Tools used
 - ii. How they were used
 - iii. Why they were chosen for that particular problem space
 - iv. Code snippets and explaining the logic behind the code snippets
 - e. Describing the 5 Insights gathered from data - 7.5 points (1.5 per each insight)
 - f. Future Work (what are the next steps for this work) - 1
 - g. References (-1 if not present)
- 3. Final Presentation (~6-8 slides) (10 points)**
- a. Each team will have 18+2 min
 - b. Presentation should contain following information
 - i. Description of the problem
 - ii. Work Distribution among team members
 - iii. Data sources with information
 - iv. Description of the processing/analysis and the tools that were used
 - v. Walk-through your analysis and your insights
 - vi. Lessons Learned
 - c. I will time and stop you at 18 min on the dot.
 - d. At the end of 18 min, I will open it out to the class for 2 min Q & A.
- 4. Voting (5 points bonus)**
- a. At the end of both the presentation sessions, each class member will vote for their favorite presentation
 - b. Most votes get 5 points, 2nd place gets 4, 3rd most votes get 3 points, rest get 2 points each
 - c. You can vote for your own projects :)

Project Topics and Datasets (examples)

Note: Don't forget, you can propose your dataset as well. You need to get approval from the instructor if you want to use other datasets (see the end of this document for a list of available sources for such datasets).

Water Quality and Potability

This dataset contains water quality measurements and assessments related to potability, which is the suitability of water for human consumption. The dataset's primary objective is to provide insights into water quality parameters and assist in determining whether the water is potable or not. Each row in the dataset represents a water sample with specific attributes, and the "Potability" column indicates whether the water is suitable for consumption.

<https://www.kaggle.com/datasets/uom190346a/water-quality-and-potability>

Spotify Top Hit Playlist (2010-2022)

The data is extracted through Spotify API directly. The dataset includes information on songs/tracks (100 per year) from Top Hit playlists from 2010 to 2022 created by Spotify.

<https://www.kaggle.com/datasets/josephinelsy/spotify-top-hit-playlist-2010-2022>

Used Cars Price Prediction

Used Car Price Prediction Dataset is a comprehensive collection of automotive information extracted from the popular automotive marketplace website, <https://www.cars.com>. This dataset comprises 4,009 data points, each representing a unique vehicle listing, and includes nine distinct features providing valuable insights into the world of automobiles.

<https://www.kaggle.com/datasets/taeefnajib/used-car-price-prediction-dataset>

Airline Dataset

It encompasses flight details, passenger profiles, and preferences, aiding airlines in optimizing services. Analyzing delay and cancellation data helps enhance punctuality and passenger experiences. Additionally, regulatory authorities depend on this data to ensure safety standards and formulate aviation policies. Researchers use it to study market trends, environmental impacts, and sustainable growth strategies. In summary, airline data is the cornerstone of informed decision-making, operational excellence, and the progression of the aviation sector.

This dataset comprises diverse parameters relating to airline operations on a global scale. These columns collectively provide comprehensive insights into passenger demographics, travel details, flight routes, crew information, and flight statuses.

<https://www.kaggle.com/datasets/iamsouravbanerjee/airline-dataset>

Cancer Database

Cancer DB is a public domain database and website containing structured data on key concepts in cancer integrated into one model. Someone's new treatment could be just 2 insights away from being the next great cure. Cancer DB can help them figure out what those 2 missing insights are.

<https://www.kaggle.com/datasets/sujaykapadnis/cancer-database>

Smoking and Drinking Dataset with Body Signal

This dataset is collected from the National Health Insurance Service in Korea. It contains the body signals and people's health status and information of being a drinker or smoker or not.

<https://www.kaggle.com/datasets/sooyoungheer/smoking-drinking-dataset>

E-commerce Customer Data For Behavior Analysis

The "E-commerce Customer Behavior and Purchase Dataset" is a synthetic dataset generated using the Faker Python library. It simulates a comprehensive e-commerce environment,

capturing various aspects of customer behavior and purchase history within a digital marketplace.

<https://www.kaggle.com/datasets/shriyashjagtap/e-commerce-customer-for-behavior-analysis>

National Health and Nutrition Examination Survey

NHANES, a program by the National Center for Health Statistics, assesses the health and nutrition status of U.S. adults and children through interviews and physical examinations. Starting in the 1960s, it transitioned to a continuous program in 1999. Each year, NHANES examines a nationally representative sample of approximately 5,000 individuals across 15 counties. The survey covers demographics, socioeconomic factors, dietary habits, and health-related questions in interviews. Additionally, it includes medical, dental, physiological measurements, and lab tests conducted by trained medical staff. NHANES plays a vital role in tracking and addressing evolving health and nutrition priorities in the United States.

<https://www.kaggle.com/datasets/cdc/national-health-and-nutrition-examination-survey>

Amazon Sales Dataset

The Dataset Contains 1K+ Amazon Product's Ratings and Reviews

<https://www.kaggle.com/datasets/ahmedsayed564/amazon-sales-dataset>

Life expectancy & Socio-Economic (world bank)

Life expectancy at birth indicates the number of years a newborn infant would live if prevailing patterns of mortality at the time of its birth were to stay the same throughout its life. It is a key metric for assessing population health.

Life expectancy has burgeoned since the advent of industrialization in the early 1900s and the world average has now more than doubled to 70 years. Yet, we still see inequality in life expectancy across and within countries.

<https://www.kaggle.com/datasets/mjshri23/life-expectancy-and-socio-economic-world-bank>

LinkedIn Job Postings Dataset

This dataset contains information about job postings on LinkedIn. The data is divided into several files, each containing different aspects of the job postings

<https://www.kaggle.com/datasets/rajatraj0502/linkedin-job-2023>

World Energy Consumption

This dataset is a collection of key metrics maintained by Our World In Data. It includes data on energy consumption (primary energy, per capita, and growth rates), energy mix, electricity mix and other relevant metrics.

<https://www.kaggle.com/datasets/pralabhpoudel/world-energy-consumption>

Netflix Movies and TV Shows

They have over 8000 movies or tv shows available on their platform, as of mid-2021, they have over 200M Subscribers globally. This tabular dataset consists of listings of all the movies and tv shows available on Netflix, along with details such as - cast, directors, ratings, release year, duration, etc.

<https://www.kaggle.com/datasets/shivamb/netflix-shows>

Football Players Stats (2025-2026)

This dataset features player statistics from the 2025-2026 season across the top five European leagues, sourced from FBref. Automatically updated weekly.

<https://www.kaggle.com/datasets/hubertsidorowicz/football-players-stats-2025-2026>

Lifestyle and Sleep Patterns

This dataset shows the relationship between lifestyle choices and their effects on overall sleep health. It includes detailed information on people's physiological factors, lifestyle choices, and sleep health, including blood pressure, heart rate, physical activity, body mass index (BMI), daily steps, stress levels, sleep duration, sleep quality, and the presence of sleep disorder.

<https://www.kaggle.com/datasets/minahilfatima12328/lifestyle-and-sleep-patterns>

World Economic Dataset

Country Profiles with Economic and Geographic Dataset

<https://www.kaggle.com/datasets/ahmadfirman/world-economic-2025>

Global Superstore Data

The provided dataset consists of three related files from a global superstore:

Orders: Contains detailed records of customer orders.

Returns: Lists orders that have been returned.

People: Maps sales personnel to specific sales regions.

Together, these files can be used to analyze sales performance, profitability, shipping efficiency, and return rates across different products, customers, and geographical locations.

Notes to the group choosing this dataset: you can read the provided codes and solutions but you cannot use them nor do a similar analysis.

<https://www.kaggle.com/datasets/rohitgrewal/global-superstore-data>

EA Sports FC 25 + Real Player Data (SoFIFA Merge)

This dataset combines EA Sports FC 25 player ratings and attributes with real-life player data such as height, weight, date of birth, clubs, and national teams. Data is collected from SoFIFA

<https://www.kaggle.com/datasets/sametozturkk/ea-sports-fc-25-real-player-data-sofifa-merge>

American Community Survey

The ACS dataset offers a nuanced exploration of socioeconomic dynamics in the United States. With a focus on income, employment, education, and demographics, it provides valuable insights into citizenship, language spoken at home, marital status, disabilities, and birth quarter, offering a comprehensive understanding of the societal landscape during that period.

<https://www.kaggle.com/datasets/sirishasingla1906/american-community-survey>

Enhanced Health Insurance Claims Dataset

This dataset contains 4500 synthetic health insurance claims, meticulously generated to simulate real-world scenarios. It is designed to be highly usable and extensive, making it ideal for machine learning, data analysis, and predictive modeling tasks. The dataset includes a variety of features that are commonly found in health insurance claims, providing a rich and diverse set of data points for analysis.

<https://www.kaggle.com/datasets/leandrenash/enhanced-health-insurance-claims-dataset>

If you want to look for other available datasets, you can check the following sites (subject to approval if you select any dataset that is not listed above)

1. Wikipedia's List of Machine Learning Datasets
https://en.wikipedia.org/wiki/List_of_datasets_for_machine-learning_research
2. Kaggle Popular Datasets
<https://www.Kaggle.com>
3. Open ML
<https://www.openml.org/search?type=data&status=active>
4. UC Irvine Machine Learning Laboratory
<https://archive.ics.uci.edu/>